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PYTHON LAB PROJECT REPORT

TITLE: Intelligent Disaster Data Interoperability and Emergency Response Coordination Platform

BY TEAM

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UNDER GUIDANCE OF

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Member 1: Disaster Data Integration and Emergency Response Coordination System (23MID0061)

Member 2 : Intelligent Disaster Assessment and Resource Management System (23MID0233)

1. MY CONTRIBUTION

Intelligent Disaster Assessment and Resource Management System

2. DESCRIPTION

The website intelligently assesses disaster situations by analyzing incident severity, shelter capacity, resource availability, and unusual utilization patterns.

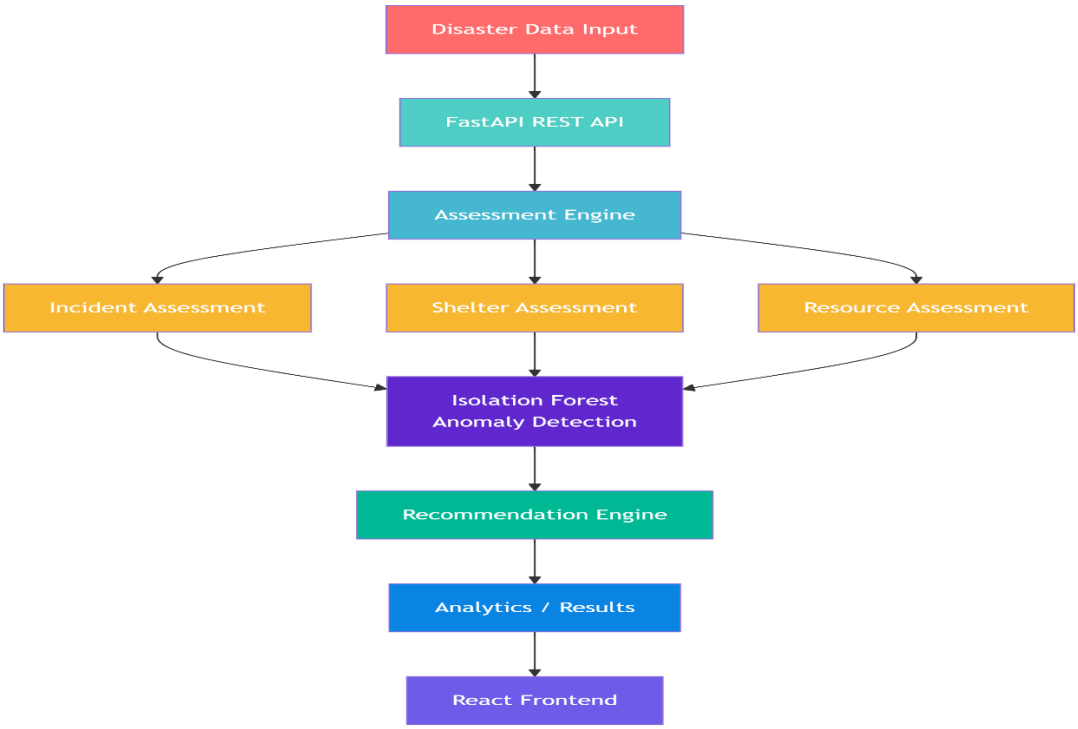
It provides priority-based assessment, ML-based anomaly detection, and actionable recommendations to support faster and better emergency resource management.

3. OBJECTIVES

The main objectives of the system are:

- To assess the severity and priority of disaster incidents.
- To monitor shelter capacity and utilization.
- To identify shortages in emergency resources.
- To detect unusual shelter utilization patterns using Isolation Forest.
- To generate priority-based disaster response recommendations.
- To provide analytical information through an interactive dashboard.
- To integrate the backend assessment system with the React frontend.

4. SYSTEM ARCHITECTURE

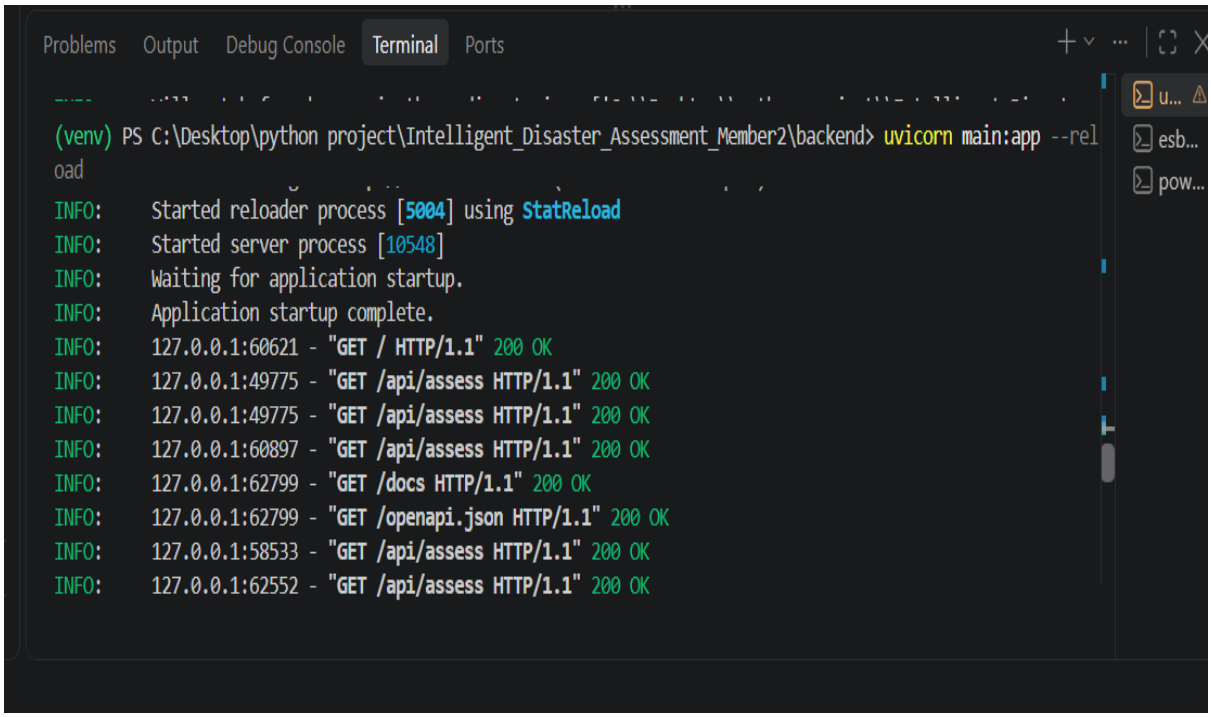


5. TECHNOLOGIES USED

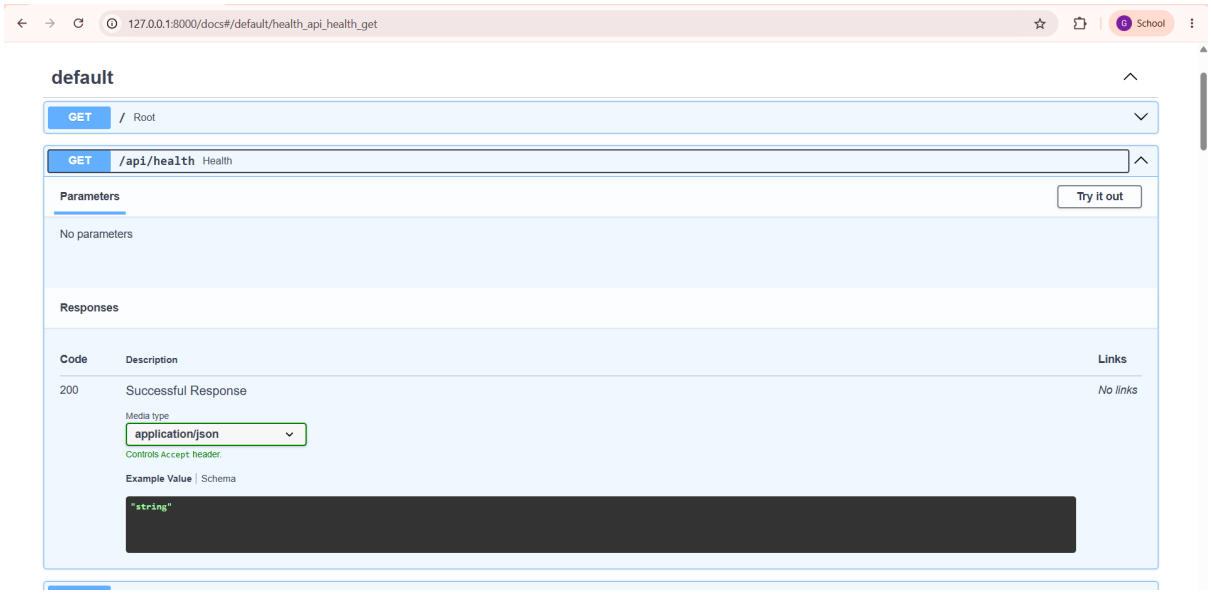
Component	Technology
Frontend	React.js
Backend	FastAPI
Programming Language	Python
API Server	Uvicorn
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn
ML Algorithm	Isolation Forest
Frontend Development	Vite
API Communication	REST API

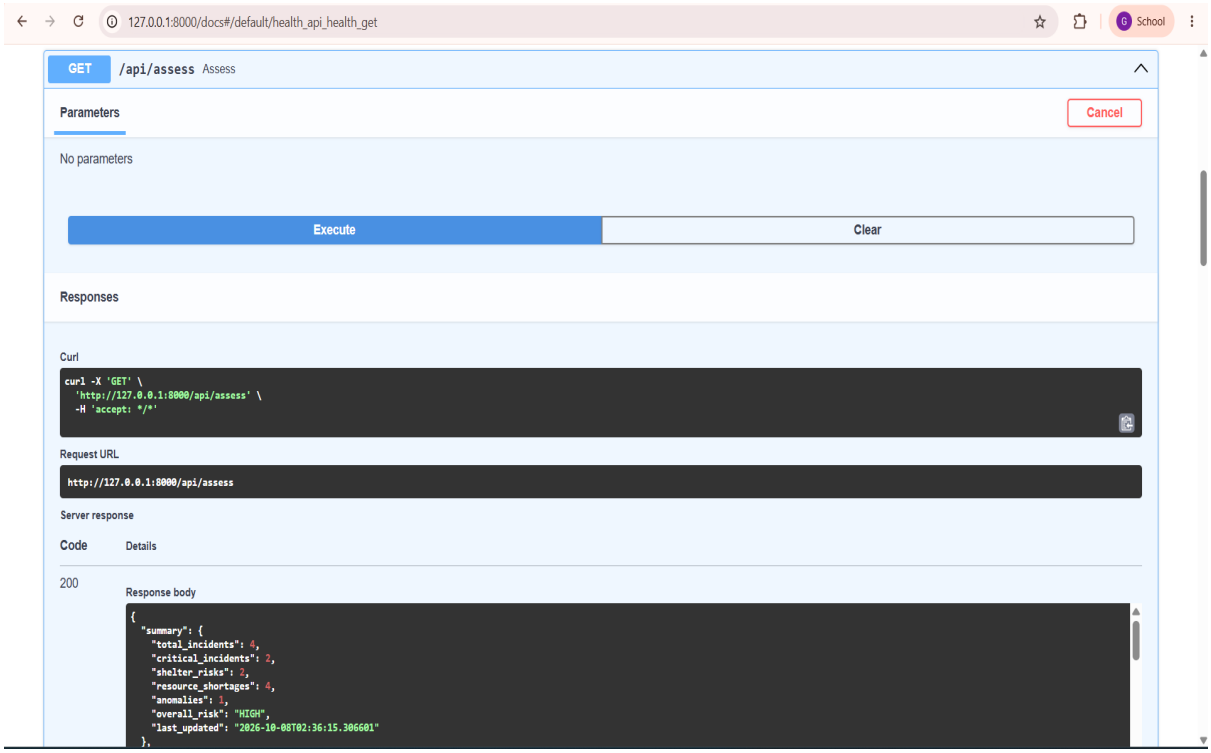
6. BACKEND IMPLEMENTATION

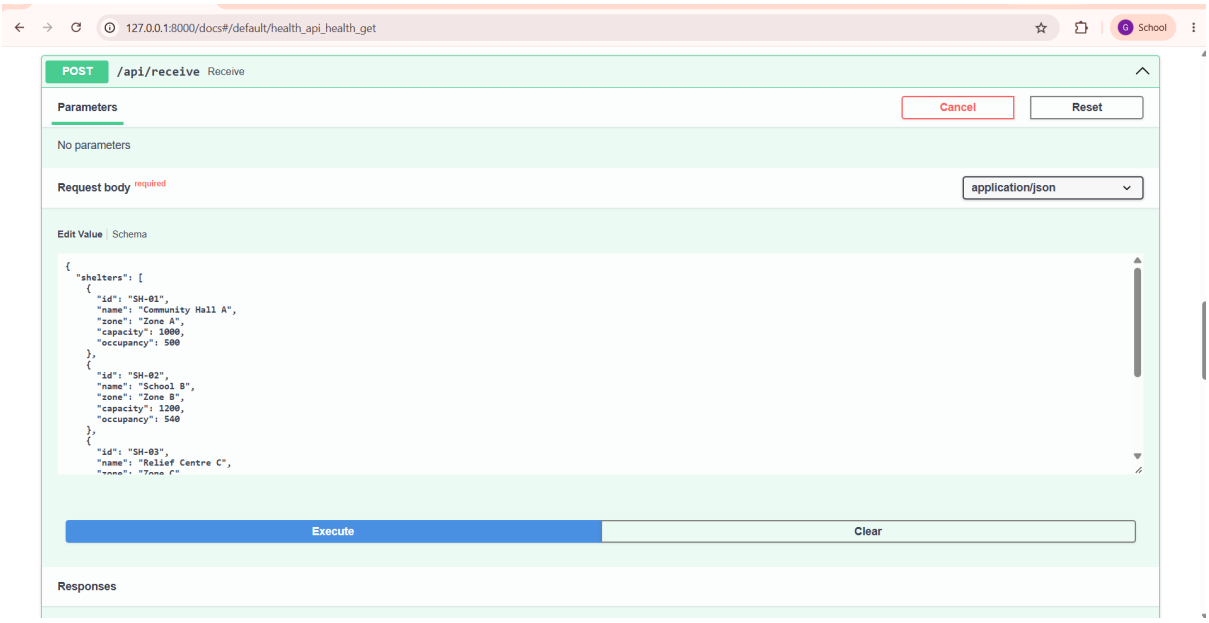
6.1 FastAPI Backend Server Running Locally

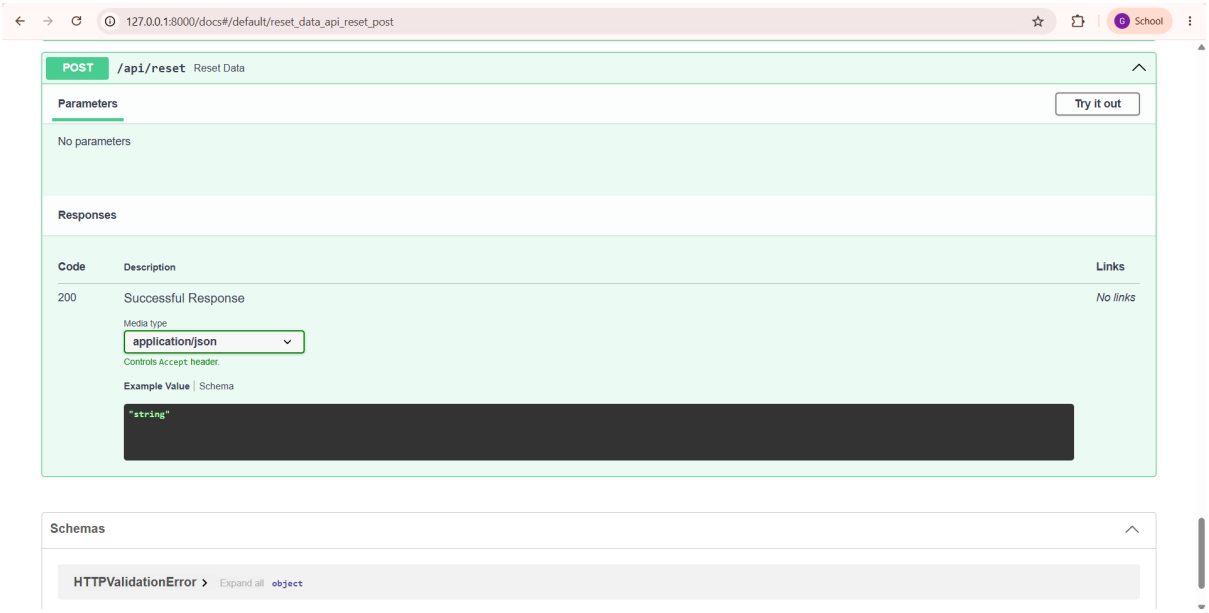


6.2 FastAPI Backend









Main API Endpoints

GET /api/health

GET /api/assess

POST /api/receive

POST /api/reset

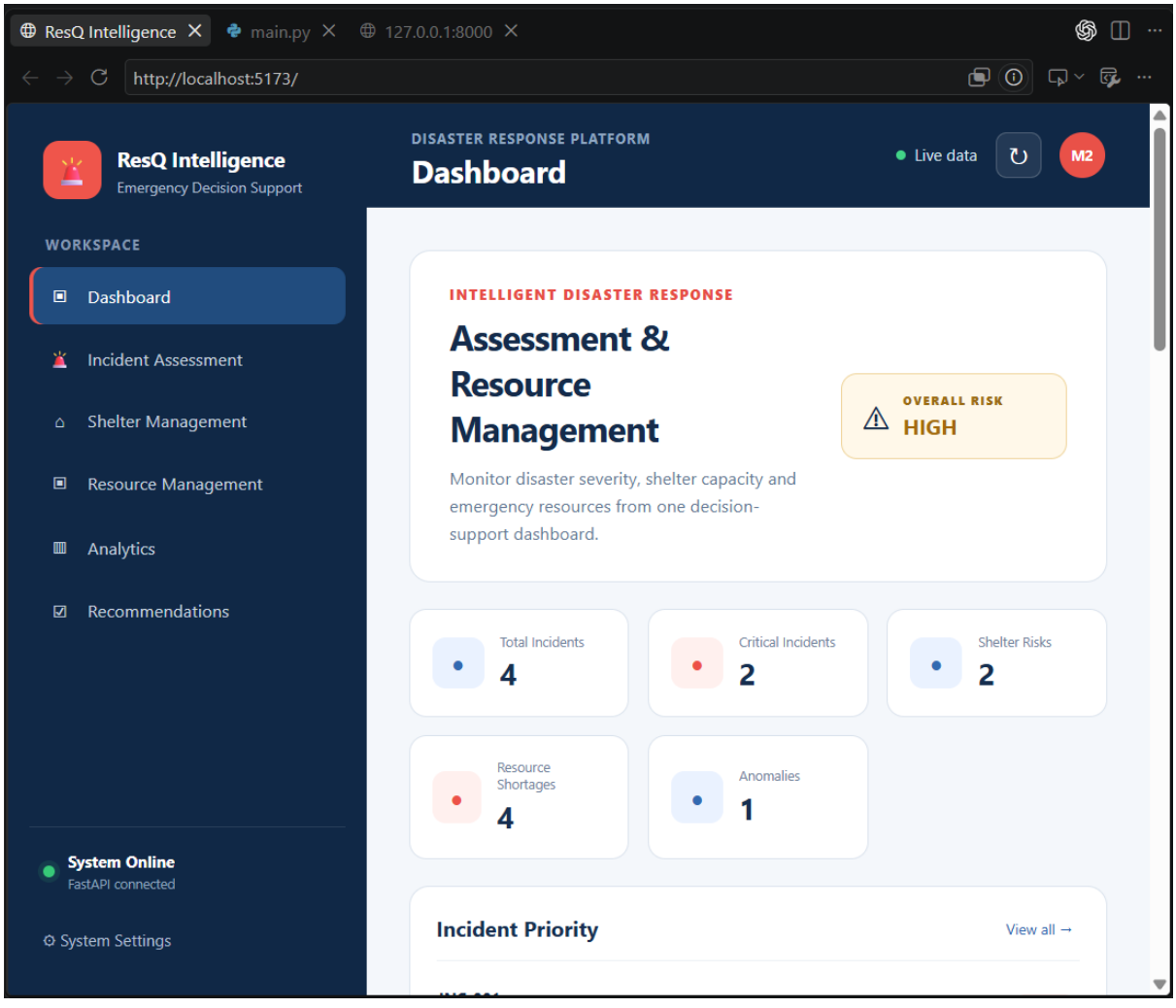
7. FRONTEND IMPLEMENTATION

The frontend was developed using **React.js** and provides an interactive dashboard for viewing disaster assessment results.

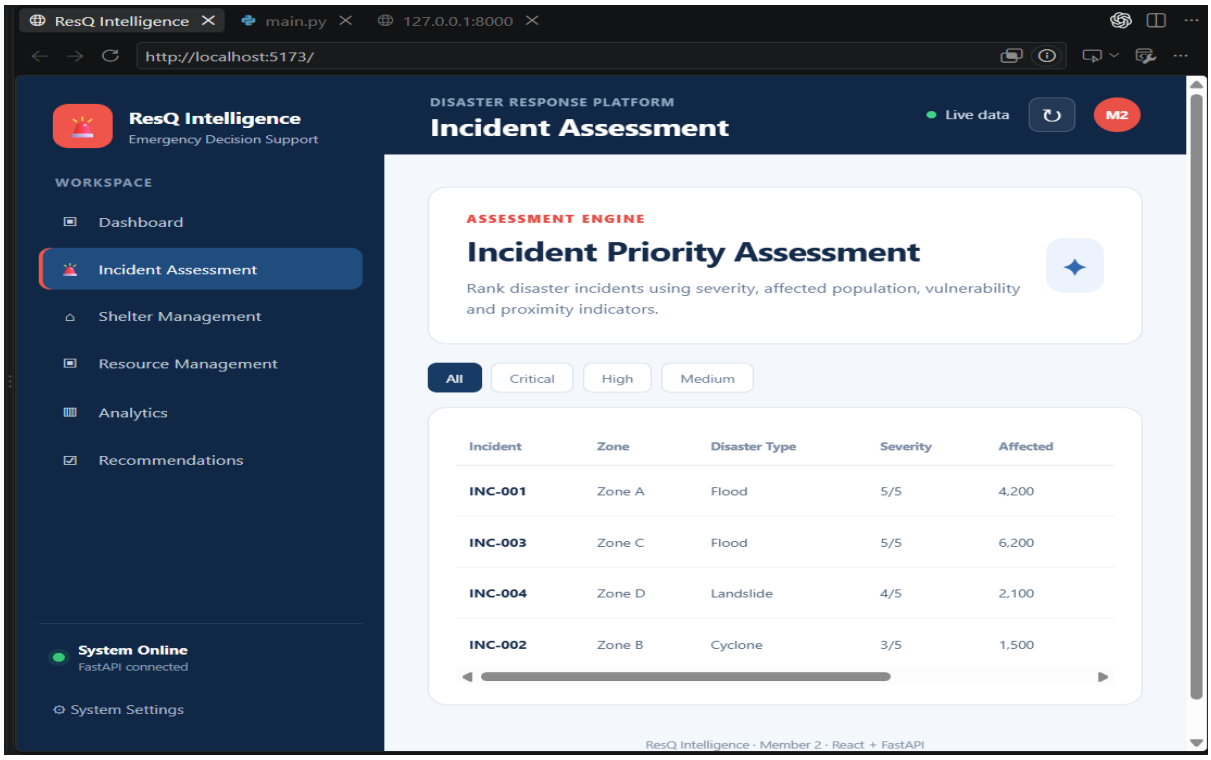
Main Modules

1. Dashboard
2. Incident Assessment
3. Shelter Management
4. Resource Management
5. Analytics
6. Recommendations

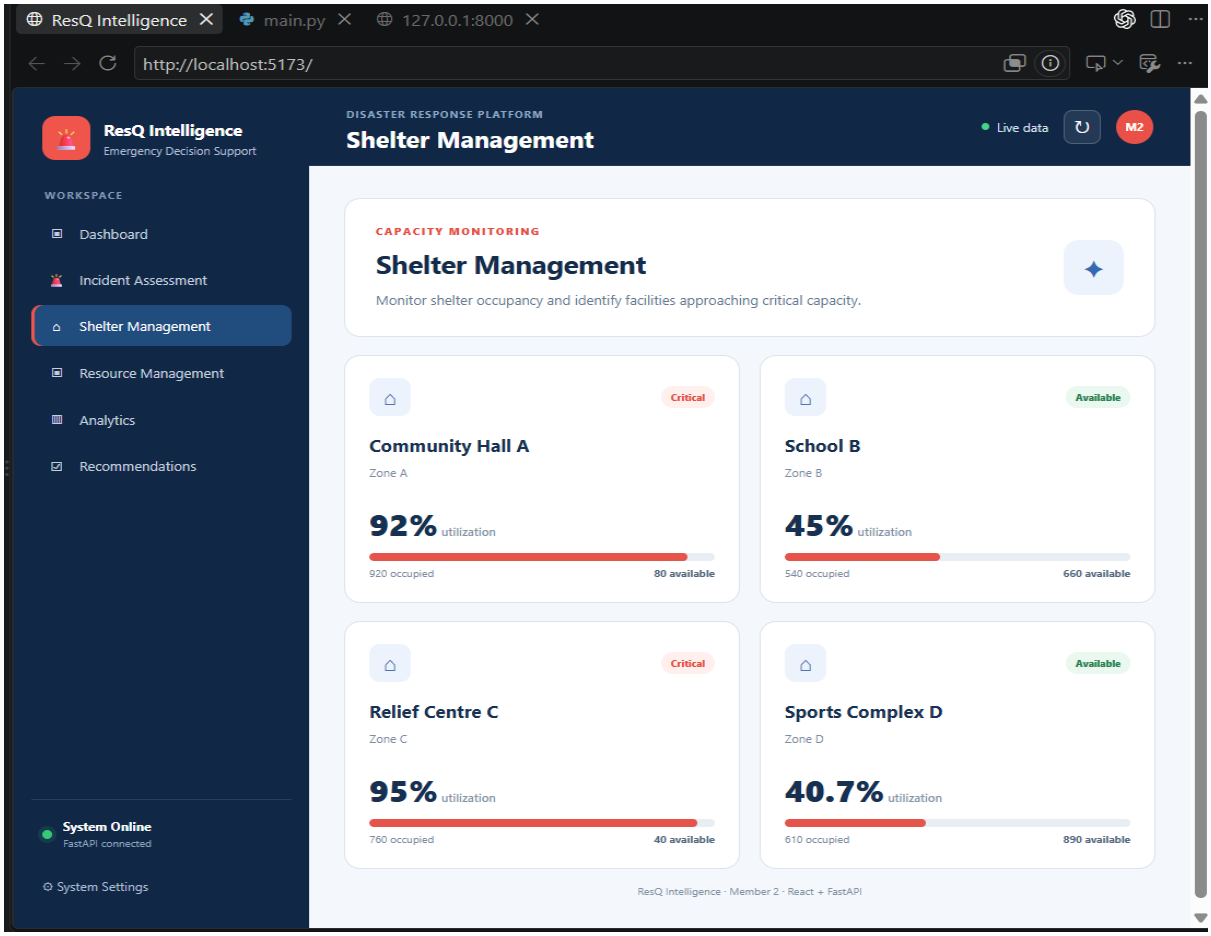
7.1. DASHBOARD



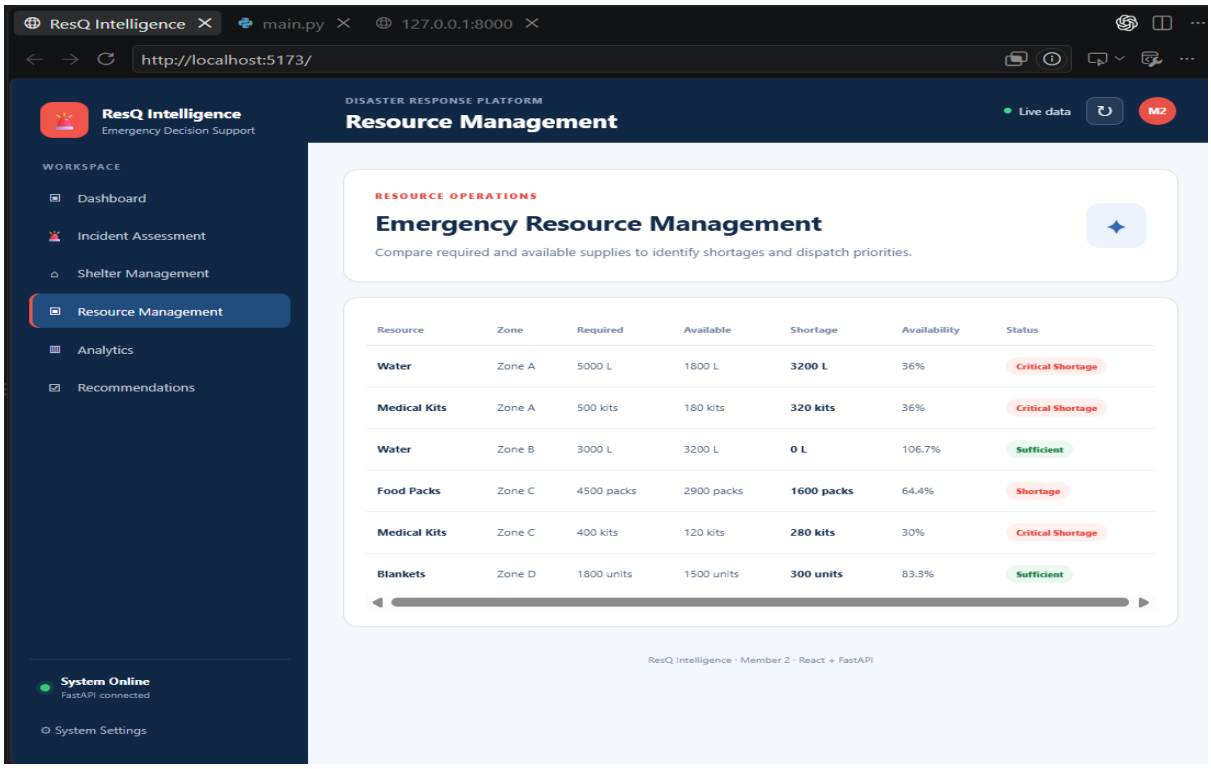
7.2 Incident Assessment



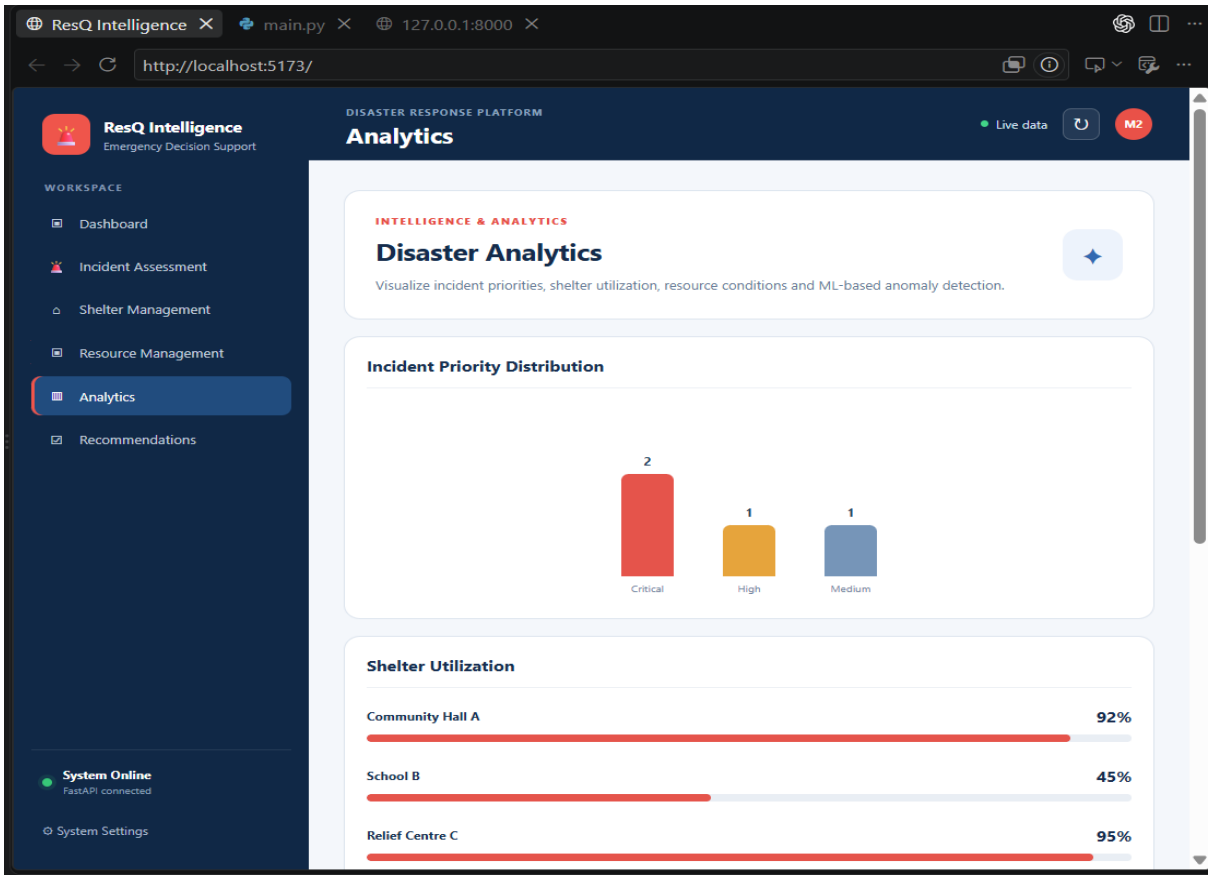
7.3 Shelter Management



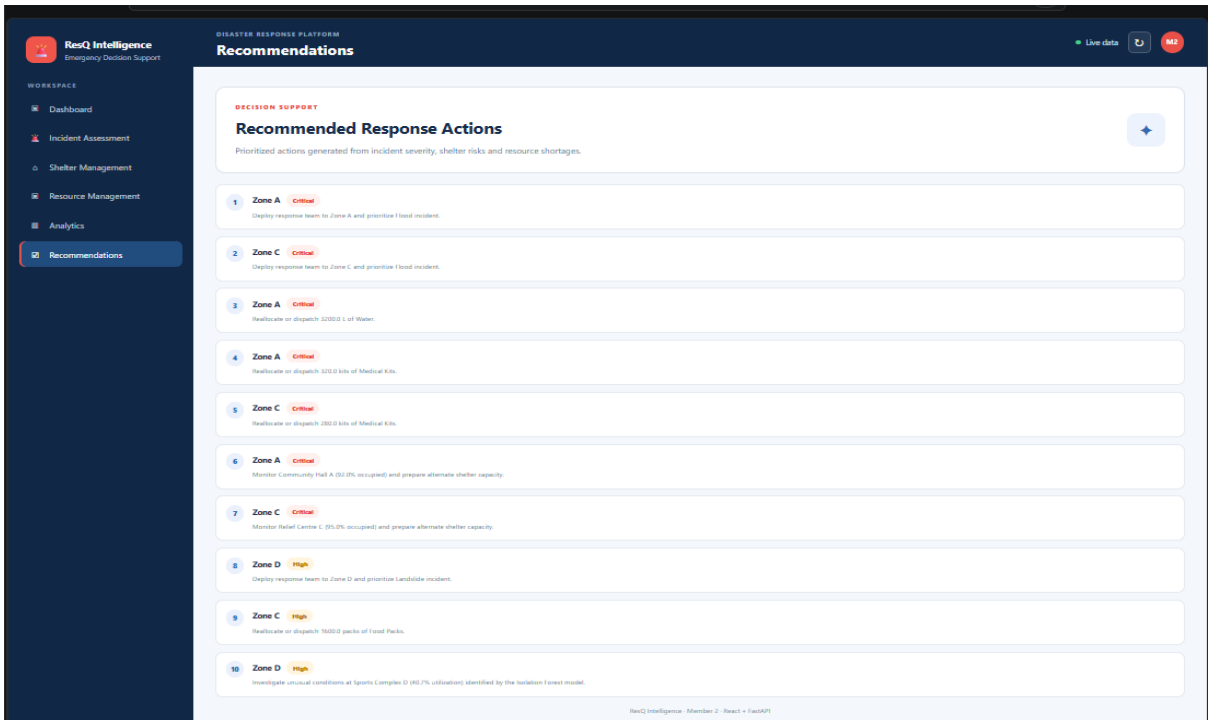
7.4 Resource Management



7.5 Analytics



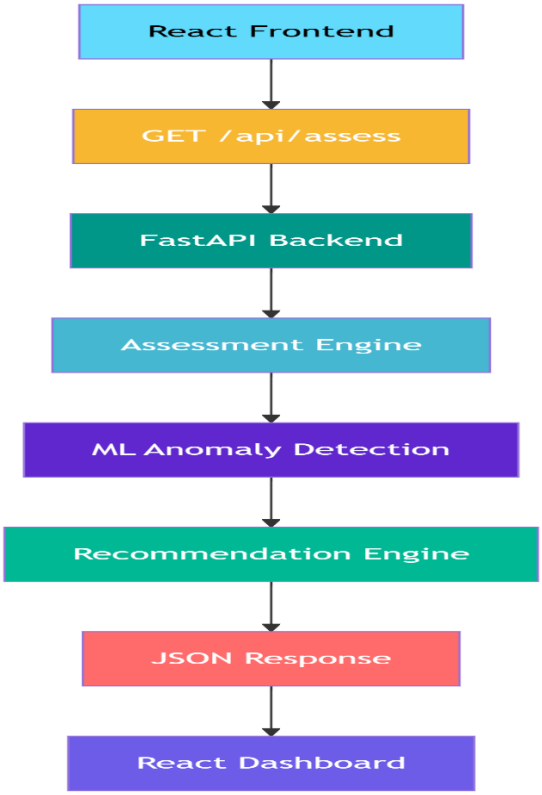
7.6 Recommendations



8. FRONTEND–BACKEND INTEGRATION

The React frontend communicates with the FastAPI backend through REST APIs.

Data Flow



9. DYNAMIC API TESTING

The /api/receive endpoint was tested by changing shelter data.

For example, Community Hall A was changed from:

920/1000 → 500/1000

The system dynamically recalculated:

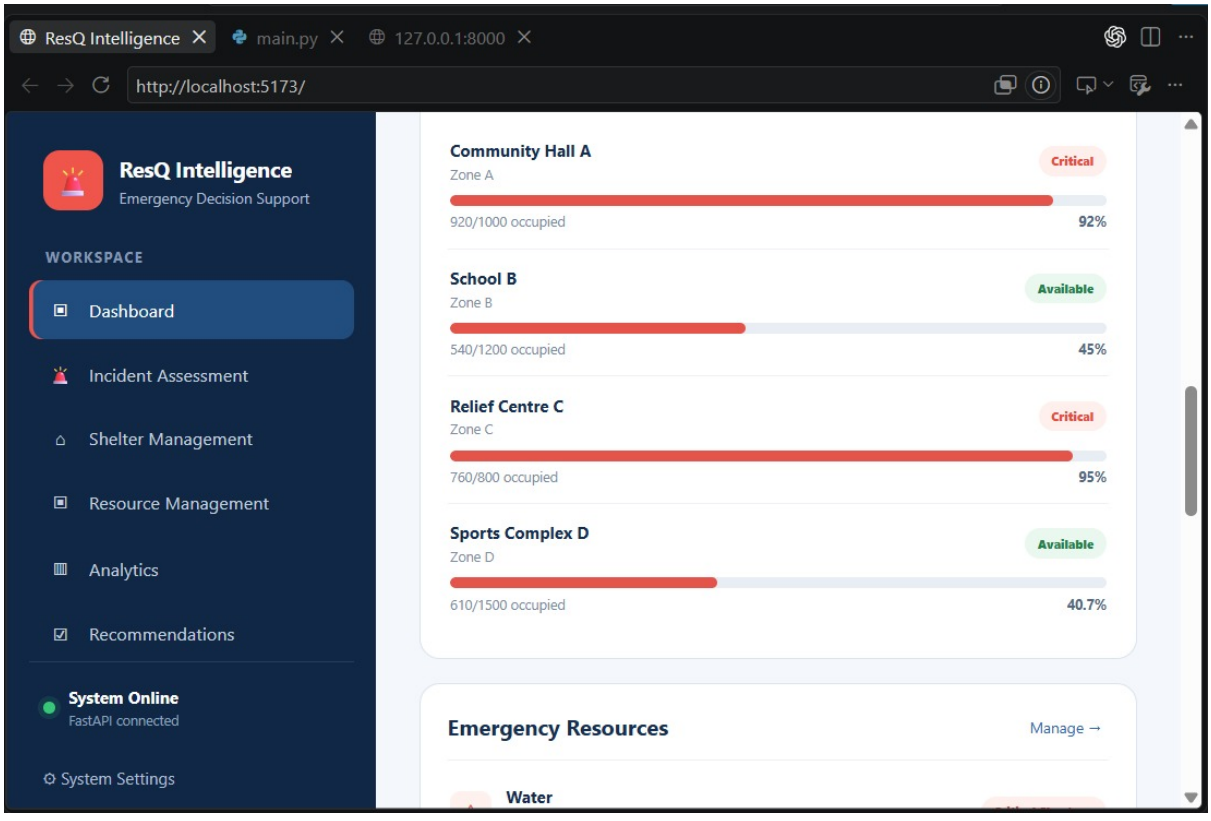
- Shelter utilization
- Shelter status
- ML anomaly results
- Recommendations
- Dashboard information

Test Result

Community Hall A changed to:

50% utilization → Available

The ML anomaly result also changed dynamically, demonstrating that the backend assessment is responsive to updated input data.



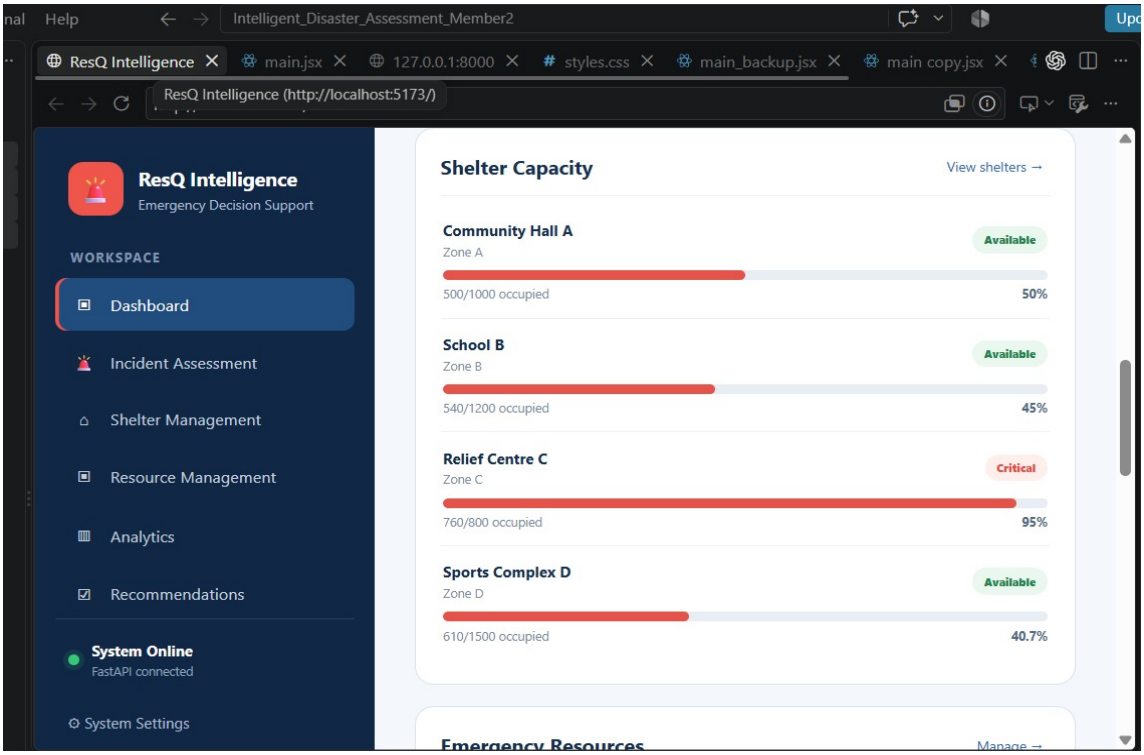


Figure : Dynamic Disaster Assessment Update Through API

10. HOSTING

Functionality	Status
Incident Priority Assessment	✓ Implemented
Shelter Capacity Assessment	✓ Implemented
Resource Shortage Assessment	✓ Implemented
ML Anomaly Detection	✓ Implemented
Recommendation Engine	✓ Implemented
Analytics Dashboard	✓ Implemented
REST API Integration	✓ Implemented
Dynamic Data Update	✓ Tested
React–FastAPI Integration	✓ Implemented

Hosting Status: Currently not hosted online.

The application is executed locally using:

- **React/Vite** for the frontend
- **FastAPI/Uvicorn** for the backend

Frontend:

http://localhost:5173/

Backend:

http://127.0.0.1:8000/

11. CONCLUSION

The Intelligent Disaster Assessment and Resource Management System provides an integrated platform for assessing disaster conditions and supporting emergency decision-making.

The system combines incident assessment, shelter capacity monitoring, resource shortage detection, Isolation Forest-based anomaly detection, and automated recommendations in a single dashboard.

The successful integration of React and FastAPI enables dynamic assessment and real-time visualization of updated disaster information.